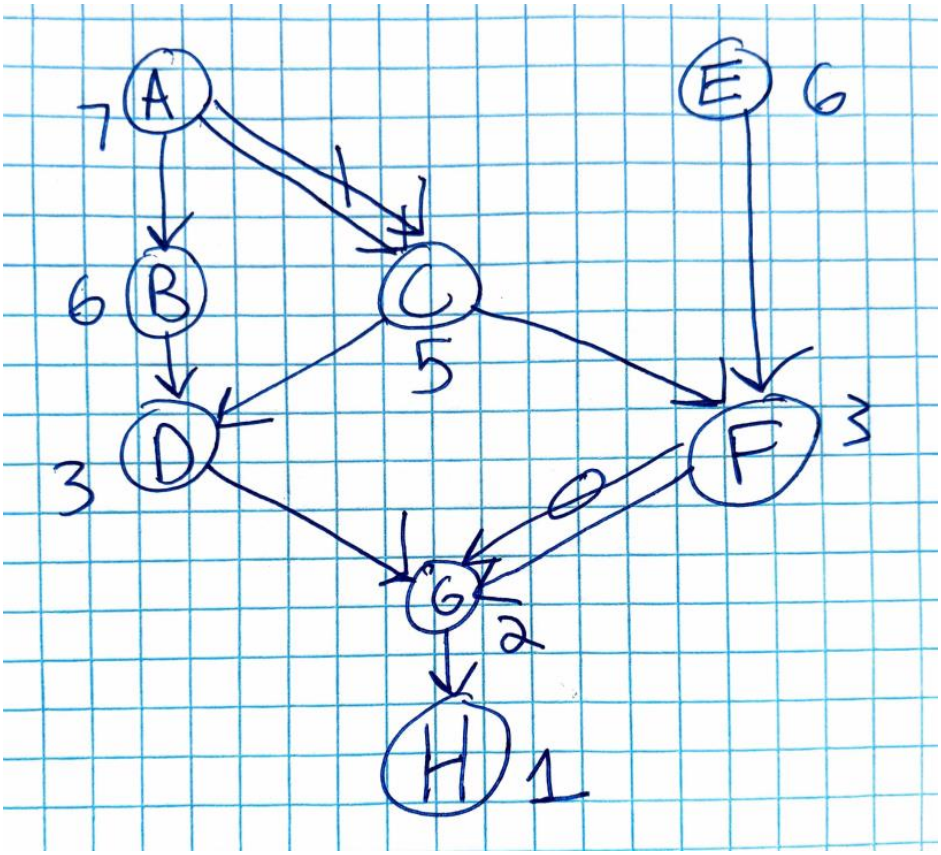
Question 1

a/b)



c)

Height Ordered
A
B
E
C
D
F
G
H

VLIW Schedule

Cycle	ALU	ALU	MUL	L/S
0	A		E	
1			B	C
2				
3	F			
4	D			
5	G			
6				H

d)

Inst	Tail	Pause
A	0	0
E	1	0
B	0	0
C	1	1
F	1	0
D	1	0
G	1	0
H	1	0

Q2) Consider the following loop. (Although it's not relevant to the solution, you can assume that R3 begins with a negative value stored in it.)

LABEL: A: ADD R1, R1, #1

B: MUL R1, R1, 4

C: LW R2, 100(R1)

D: SW R2, 200(R1) // Note: R2 is a SOURCE register for SW

E: ADD R3, R3, #1

F: BRNEZ R3, LABEL // branch to LABEL if R3 != 0

Referring to Software Pipelining via Iterative Modulo Scheduling, what is the RecMII and ResMII for this loop? What is the overall minimum II? Assume there is 1 ALU unit, 1 MUL unit, 1 LW unit and 1 SW unit. Both MUL and LW takes 2 cycles and are pipelined. SW, ADD and BREZ all take 1 cycle. ADD and BREZ use the ALU unit. Now show a prologue, kernel and epilog after modulo scheduling. For simplicity, you can write the letter of the operation in a table. You do not need to write the instructions themselves.

1) ResMII

		ALU	MUL	LW	SW
Available EX unit		1	1	1	1
Instruction count		3	1	1	1
Latency		1	2	2	1
Instr_count/#EX unit		3	1	1	1

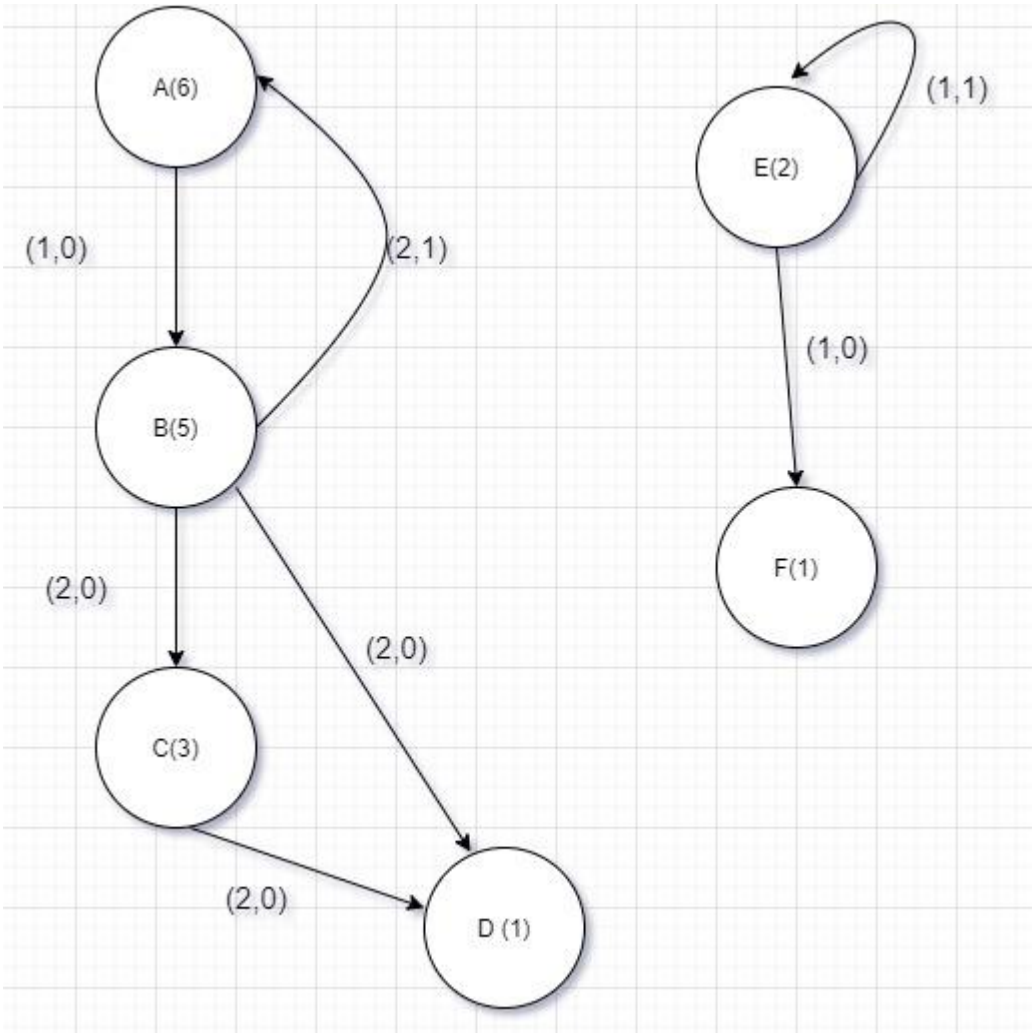
ResMII = Maximum of Instruction count/Available function unit for all functional units

= 3 (for ALU)

ResMII = 3

2) RecMII

Data dependence graph:



Cycle 1 = A-> B -> A

Cycle length = 1 + 2 = 3

Iteration distance = 0 + 1 = 1

Cycle length/iteration distance = 3

Cycle 2 = E-> E

Cycle length = 1

Iteration distance = 1

Cycle length/iteration distance = 1

RecMII = Max (3, 1) = 3

MinII = Max (ResMII, RecMII) = 3

Modulo scheduling for a given branch behaviour (T,T,T,NT):

Branch behaviour	ALU	MUL	LW	SW
	A0			
	E0	B0		
Branch taken	F0			
	A1		C0	
	E1	B1		
Branch taken	F1			D0
	A2		C1	
	E2	B2		
Branch taken	F2			D1
	A3		C2	
	E3	B3		
Branch not taken	F3			D2
			C3	
				D3

From above scheduling:

		ALU	MUL	LW	SW
		A			
Prologue		E	B		
		F*			

F* is branch instruction with condition opposite to F and go to epilog if condition is true.

		ALU	MUL	LW	SW
		A		C	
Kernel		E	B		
		F			D

		ALU	MUL	LW	SW
				C	
Epilog					
					D